

Literature Benchmark Brief: Ravikumar_2020_TSG

Title: Anomaly detection and mitigation for wide-area damping control using machine learning

Authors: G. Ravikumar, M. Govindarasu (2020)

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Cluster: Cluster B: SCADA Degradation & Reliability

Core Theoretical Mechanism & Model Formulation

Multi-class anomaly detection, wide-area/local dual-mode switching, convex mitigation

Empirical Scale & Experimental Dataset

IEEE 39-bus and 68-bus Real-Time Digital Simulator (RTDS)

Relevance to IEEE TSTE & Sustainable Energy Systems

Demonstrates that ML models fail when fed corrupt inputs, requiring explicit fallback

Identified Gaps & Operational Vulnerabilities

Does not address physical control cliff near aerodynamic saturation

Role in Manuscript Defense & Reviewer #2 Proof

Cited in Section I and IV to prove that unhedged ML under corrupted inputs is dangerous